

Ultimate Nutrition

Beta Alanine

The use of Beta-Alanine to increase muscle Carnosine offers a way to increase anaerobic exercise potential and, is likely to add to the effects of Creatine in some exercise settings.

First discovered in the early 1900's, Beta-Alanine and Histidine are the two components of Carnosine. Histidine is already present in large quantity within skeletal muscles, so it is Beta-Alanine that acts as the rate-limiting factor in Carnosine conversion.

A buildup of metabolic waste limits muscle contraction, with Hydrogen ions being the worst. This is primarily true of our fast fibers, which are most sensitive to Hydrogen ions and extremely susceptible to fatigue. If the Hydrogen ions can be buffered, muscle strength can be maintained for a longer time before fatigue sets in. This means that anyone involved in exercise where lactic acid buildup is the limiting factor, including bodybuilders, stands to benefit from beta-alanine. In the gym, this may translate into more reps with a given weight.

Carnosine is very effective at buffering the Hydrogen ions responsible for producing the lactic acid burn. Studies have shown that increasing muscle Carnosine by supplementing Beta-Alanine may delay fatigue and improve the muscular aspects of athletic performance.

Research also indicates that Beta-Alanine increases lactate threshold, improves the ability to maintain maximal power output during high-intensity exercise and decreases neuromuscular fatigue. By elevating Carnosine level, it may protect against damage to nerves, letting them fire at a faster rate than if they are injured. For example, instead of ones body operating at 80% the day after exercise, Carnosine may help it perform closer to the best possible levels. This is mainly useful for athletes who are constantly using the same muscles, without the ability to take a break and recover for a few days.

One study compared orally administered Beta-Alanine in 10, 20, and 40 mg/kg levels to humans, and found that less than 5% was excreted in urine, and that in the above amounts, increases in muscle Carnosine stores were 42.1, 64.2 and 65.8%, respectively. In addition, workout volume increased by 13% after four weeks, and 16.2% after 10 weeks. This suggests that for a 220 lb athlete, 4g per day is more than enough. Real world feedback, in addition to dozens of human studies on strength, endurance, strength/endurance, and power/endurance athletes shows increases in muscular strength, size, workout volume, and general work capacity of all three types of muscle fibers.

SELECTED REFERENCES:

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7. T. Nagasawa Et. Al. In vitro and in vivo inhibition of muscle lipid and protein oxidation by carnosine. *Molecular and Cellular Biochemistry*. September 2001.

FREQUENTLY ASKED QUESTIONS:

1. Shouldn’t we take extra histidine along with Beta-Alanine since histidine is a component of carnosine?

No, because histidine is already present in high concentrations in muscle, while Beta-Alanine is only present only in small amounts. Researchers have determined that it is beta-alanine that drives carnosine synthesis, not histidine. For the majority of healthy people, eating a well balance diet, histidine deficiency is very uncommon and no extra is needed to increase carnosine, just Beta-Alanine.

2. Does Beta-Alanine replace creatine?

Beta-Alanine does not replace creatine. As shown above, they work differently and creatine is still effective for increasing maximal strength and power. If anything, they should be taken together as the ultimate one-two punch.

3. What is the tingling feeling I am experiencing?

Beta-Alanine is known to cause a niacin-like skin tingling reaction within minutes after it is consumed. This is perfectly normal and lessens with continued Beta-Alanine supplementation. It has best been described as the skin having begun to tingle or itch in certain spots. This is a normal body reaction to Beta-Alanine consumption.